Forecasting Airport Congestion at Madrid-Barajas

Using flight tracking, weather, and calendar signals to predict congestion pressure 1-day ahead with Eurocontrol IFR traffic data (2017--2026)

3,318	9+ years	53	$R^2 = 0.96$
DAILY RECORDS	JAN 2017 -- FEB 2026	FEATURES	TEST PERFORMANCE

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1 Project Overview

This project forecasts **airport congestion pressure** at Adolfo Suarez Madrid-Barajas (LEMD), Spain's busiest airport. We built a machine learning pipeline that ingests daily IFR flight movements from Eurocontrol, enriches them with weather data and calendar signals, computes a composite congestion score (ACPS), and trains gradient-boosted tree models for both regression and classification.

> **Key question:** Can we forecast airport congestion pressure using flight volume, weather, and calendar signals?
> **Answer:** Yes -- our model achieves R^2 = 0.96 on unseen test data, accurately predicting daily congestion with a mean absolute error of just 0.45 points on a 0--100 scale.

The original prototype used ~25 hourly records from FlightRadar24 scraping. This new pipeline uses **3,318 daily records** from the official Eurocontrol IFR traffic dataset, covering January 2017 through February 2026 -- a 130x increase in data volume.

2 Data Sources

E — Eurocontrol IFR Daily Traffic

Airport_Traffic.xlsx (DATA sheet) -- 1,021,929 rows across all European airports. Filtered to LEMD: 3,346 daily records with IFR departures, arrivals, and total movements. Network Manager (NM) figures used (zero nulls vs 29 nulls in airport-reported figures).

W — Open-Meteo Historical Weather

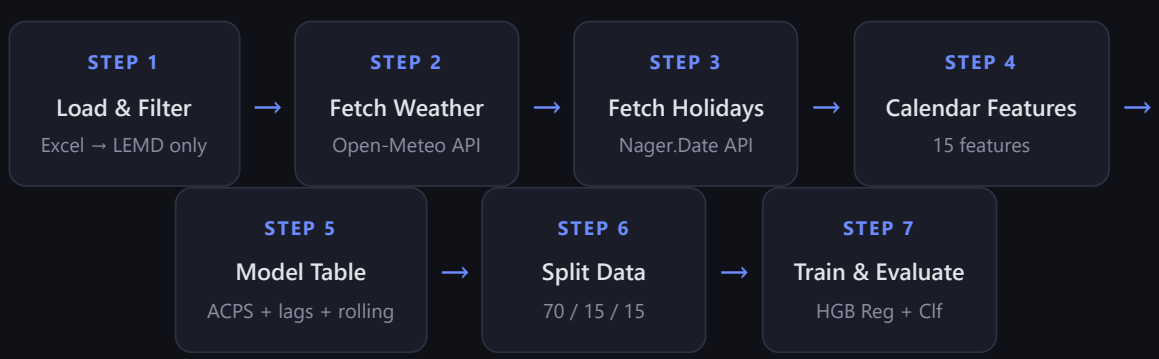
Free API providing hourly weather for Madrid-Barajas coordinates (40.47N, -3.57W). 10 variables fetched: temperature, humidity, precipitation, rain, weather code, surface pressure, wind speed/direction/gusts, cloud cover. Aggregated from hourly to daily (means, maxes, sums).

H — Nager.Date Public Holidays

Spanish public holidays for 2017--2026 (99 dates total). Used to build holiday, pre-holiday, post-holiday, and bridge-day indicator features that capture demand shifts around non-working days.

3 Processing Pipeline

The entire pipeline runs in a single script (`scripts/process_eurocontrol_data.py`) with 7 sequential steps:



Step 1: Load & Filter Eurocontrol Data

The Eurocontrol `Airport_Traffic.xlsx` file contains daily IFR movement counts for every major European airport since 2017. We read the `DATA` sheet (1,021,929 rows) and filter to `APT_ICAO == "LEMD"`, yielding 3,346 daily records. We use the **Network Manager (NM)** columns (`FLT_DEP_1`, `FLT_ARR_1`, `FLT_TOT_1`) which have zero null values, rather than the airport-reported IFR figures (29 nulls).

Step 2: Fetch Weather Data

Hourly weather is fetched from the Open-Meteo Historical Weather API for the Madrid-Barajas coordinates. The API has a ~1-year limit per request, so we chunk the 9-year range into yearly segments. The 10 raw hourly variables are aggregated to daily:

- **Mean:** temperature, humidity, wind speed, pressure, cloud cover, wind direction
- **Sum:** precipitation, rain
- **Max:** weather code (severity), wind gusts, peak wind speed

Engineered features are added: sin/cos wind direction encoding, severe weather flag (WMO code ≥ 65), and rain indicator. All weather data is cached to parquet after first fetch.

Step 3: Fetch Holidays

Spanish public holidays for each year (2017--2026) are fetched from the Nager.Date API. Only "global" holidays (national, not regional) are included. Each year's JSON response is cached locally. Total: **99 holiday dates**.

Step 4: Calendar Features

From each date, we extract 15 features:

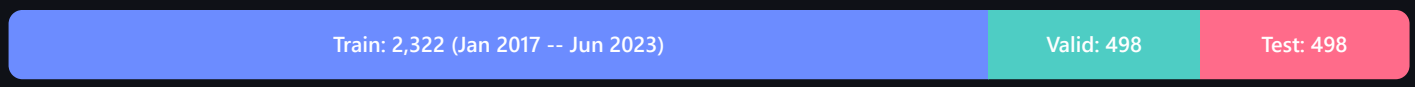
- **Temporal:** day of week (0-6), month (1-12), quarter, day of year
- **Binary:** is_weekend, is_holiday, is_pre_holiday, is_post_holiday, is_bridge_day
- **Cyclic:** sin/cos encodings of day-of-week, month, and day-of-year (to capture periodicity without artificial boundaries)

Step 5: Build Model Table

All sources are merged on date. The **ACPS target** (Airport Congestion Pressure Score, 0--100 scale) is computed, along with lag features, rolling windows, and additional derived features. See Section 5 for details.

Step 6: Chronological Split

The data is split chronologically (no shuffling, to prevent data leakage from the future):



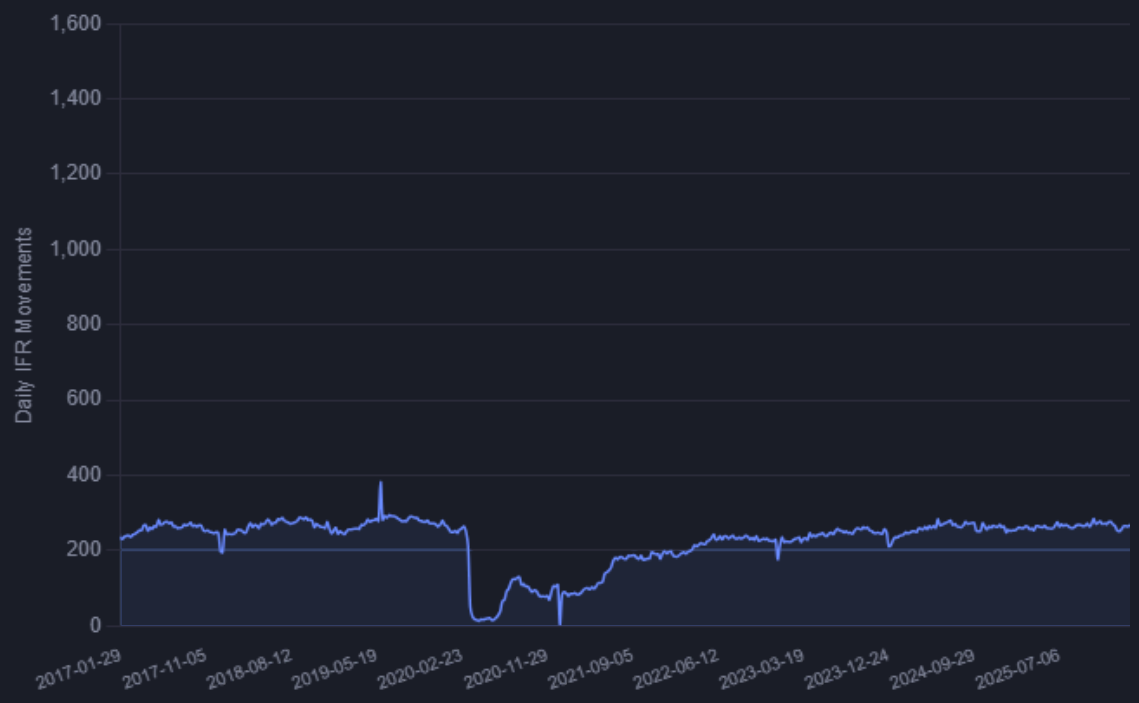
Step 7: Train & Evaluate

A **HistGradientBoostingRegressor** predicts continuous ACPS. A **HistGradientBoostingClassifier** predicts congestion level (Low/Medium/High). Both use max_iter=200, max_depth=6, learning_rate=0.1. Evaluation uses MAE, RMSE, R^2 for regression and accuracy/F1/balanced accuracy for classification.

4 Exploratory Data Analysis

Daily Flight Movements (2017--2026)

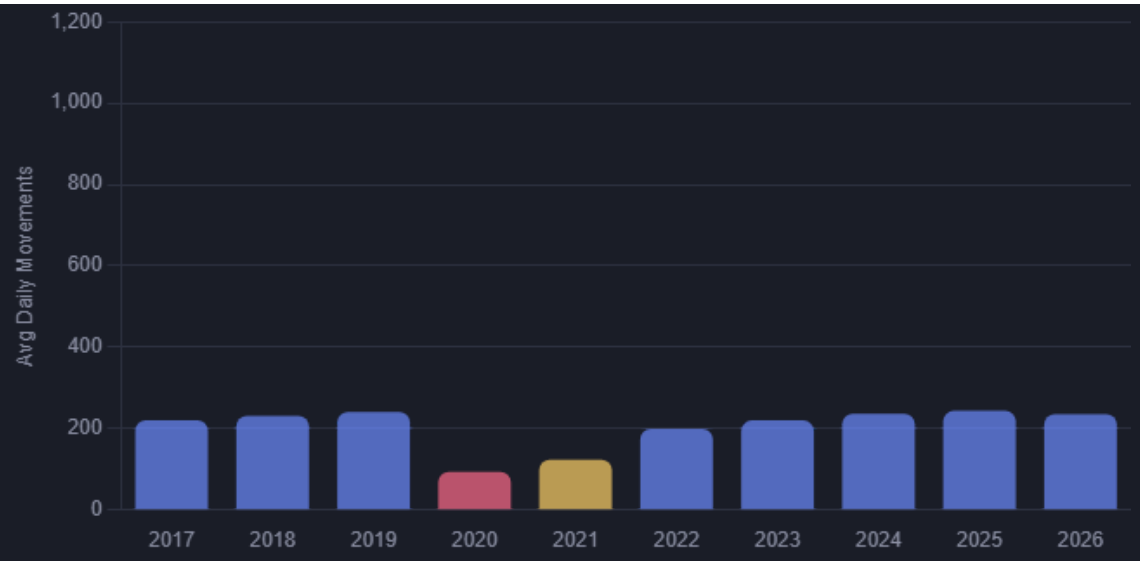
TOTAL IFR MOVEMENTS PER DAY AT MADRID-BARAJAS



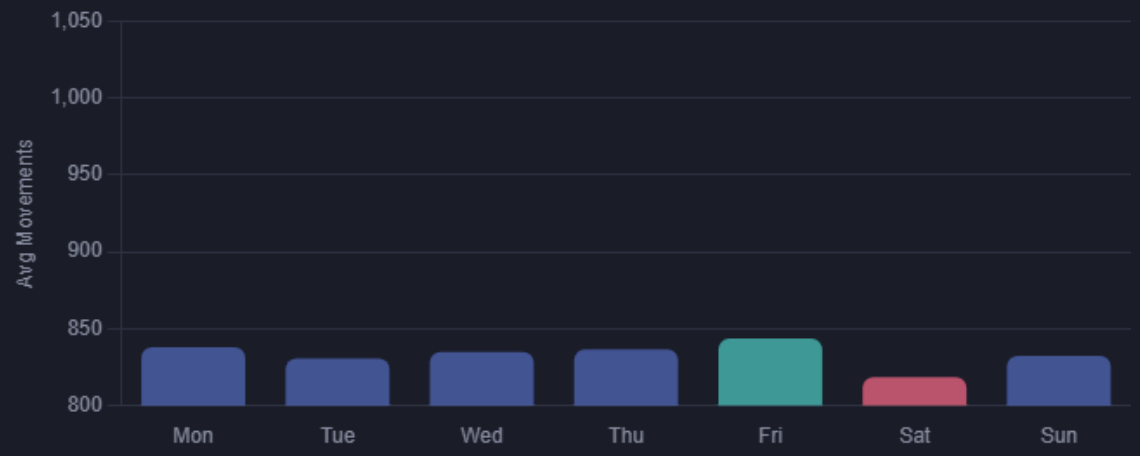
The time series reveals three distinct phases: **pre-COVID stability** (2017--2019, ~1,000--1,200 daily movements), the **COVID crash** (March 2020, dropping to near-zero), and the **gradual recovery** (2021--2023, returning to pre-pandemic levels by mid-2023). By 2025, traffic slightly exceeds 2019 levels.

Yearly Averages

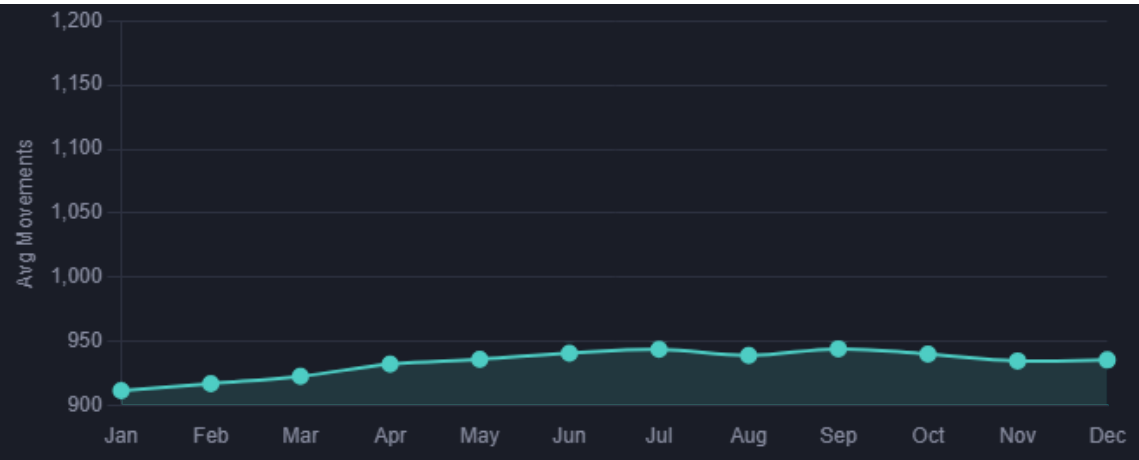
AVERAGE DAILY MOVEMENTS BY YEAR



DAY-OF-WEEK PATTERN



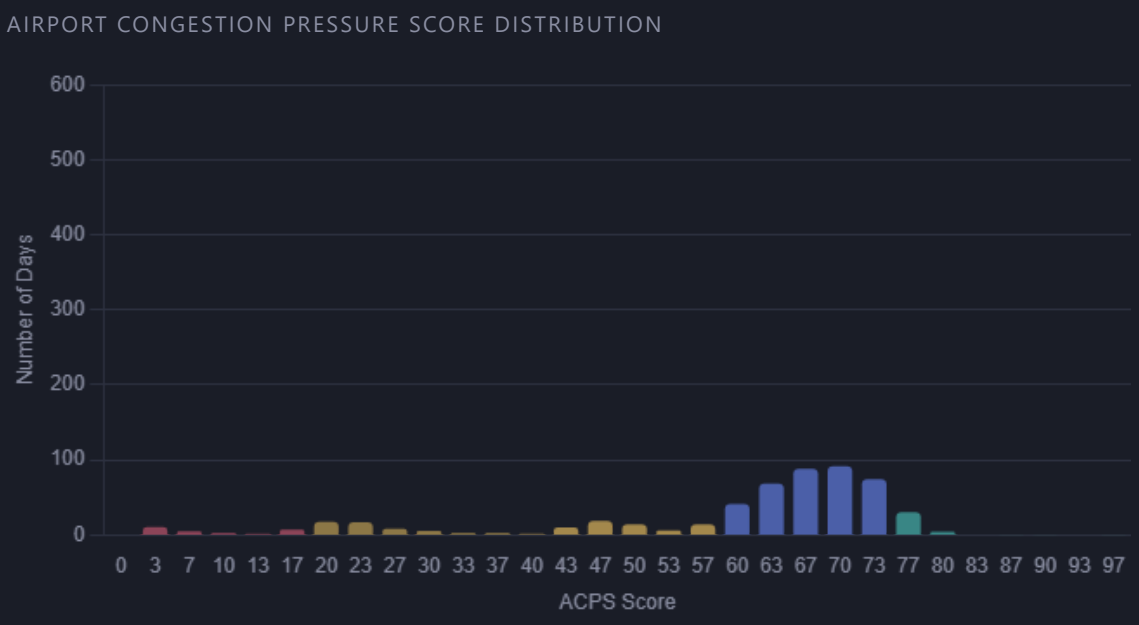
MONTHLY SEASONALITY (2022--2025)



Day-of-week: Saturday is consistently the quietest day (~898 avg), while Friday is the busiest (~1,030). This reflects business travel patterns concentrated on weekdays.

Seasonality: June--September shows peak traffic (~1,130--1,150 daily), driven by summer tourism. January is the quietest month (~966). This summer peak is critical context for congestion forecasting.

ACPS Distribution



The ACPS distribution is bimodal: a cluster of low scores (0--35) corresponds to the COVID period, while the main mass (60--78) represents normal operations. The three congestion classes split at the 60th and 85th percentiles.

1,980

LOW CONGESTION DAYS

59.7% of dataset

837

MEDIUM CONGESTION DAYS

25.2% of dataset

501

HIGH CONGESTION DAYS

15.1% of dataset

5 Feature Engineering

The model table contains **53 numeric features** across five categories:

CATEGORY	FEATURES	COUNT	EXAMPLES
Flight volume	Raw movement counts and balance	4	arrivals, departures, total_movements, arr_dep_imbalance
Calendar	Temporal patterns and holidays	15	dow, month, is_weekend, is_holiday, is_bridge_day, dow_sin/cos, month_sin/cos, doy_sin/cos
Weather	Daily weather conditions	11	temperature_2m, precipitation, wind_speed_10m, wind_dir_sin/cos, is_severe_weather, is_raining
Lag features	Past ACPS and movements	14	acps_lag_1d/2d/3d/7d/14d/28d/365d, movements_lag_*
Rolling windows	Trend and volatility	9	acps_rmean_7d/14d/28d, movements_rmean_*, acps_rstd_28d, movements_yoy_change

ACPS Target Computation

The Airport Congestion Pressure Score is a composite metric:

```
ACPS = rescale_to_0_100(0.6 * z(total_movements) + 0.4 * z(pressure_ratio))
```

```
where:
  z(x)            = (x - mean(x)) / std(x)          # z-score standardization
  pressure_ratio  = movements / median(movements for same day-of-week)
  rescale(x)      = (x - min) / (max - min) * 100    # normalize to 0-100 scale
```

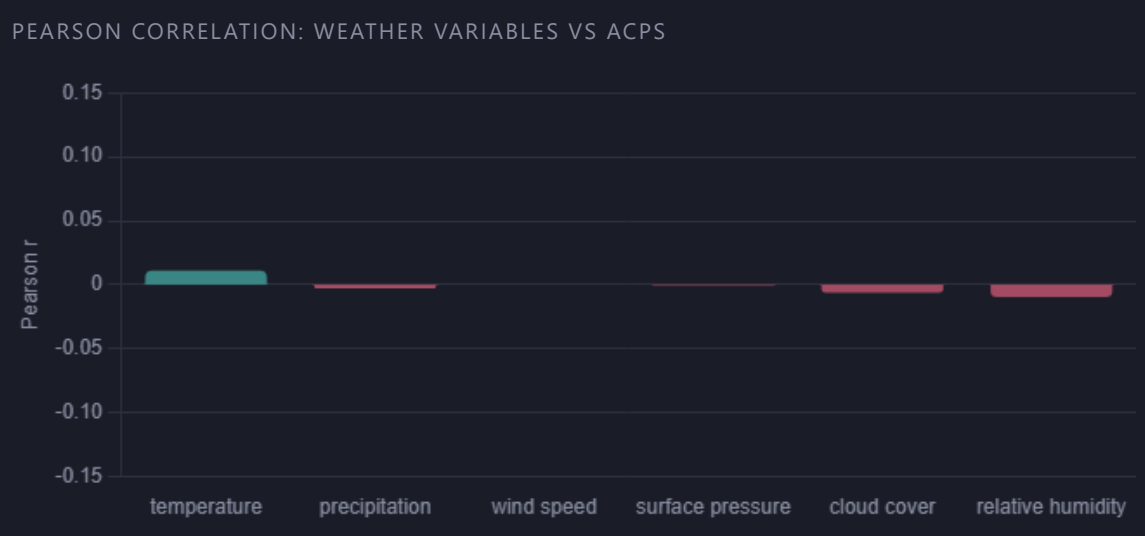
The 60/40 weighting means raw traffic volume (movements) is the primary driver, while the pressure ratio captures whether the airport is busier than typical for that day of the week. The z-score standardization ensures both components contribute proportionally regardless of their raw scales.

Congestion Classification

ACPS is discretized into three classes based on percentile thresholds:

- LOW ACPS below 60th percentile -- normal operations
- MEDIUM ACPS between 60th and 85th percentile -- elevated pressure
- HIGH ACPS above 85th percentile -- significant congestion risk

Weather Correlation with ACPS



Weather has weak direct correlation with ACPS at the daily level. Temperature shows the strongest relationship (r = 0.11), likely as a proxy for season (summer = more flights). Weather's value is more in capturing edge cases: severe storms or heavy rain days that cause operational disruptions.

6 Model Training

Algorithm: HistGradientBoosting

We use scikit-learn's **HistGradientBoostingRegressor** and **HistGradientBoostingClassifier**, which are histogram-based gradient boosting implementations similar to LightGBM. Key advantages:

- Native handling of missing values (NaN in lag_365d features)
- Fast training on medium-sized tabular data
- Built-in regularization through max_depth and min_samples_leaf
- No feature scaling required

Hyperparameters

PARAMETER	VALUE	RATIONALE
max_iter	200	Sufficient boosting rounds for 2,322 training samples
max_depth	6	Allows moderate interaction complexity
learning_rate	0.1	Standard rate balancing speed and accuracy
min_samples_leaf	20	Prevents overfitting to rare patterns
random_state	42	Reproducibility

Data Splits

SPLIT	ROWS	DATE RANGE	PURPOSE
Train	2,322 (70%)	Jan 2017 -- Jun 2023	Model fitting
Validation	498 (15%)	Jun 2023 -- Oct 2024	Hyperparameter tuning, early stopping
Test	498 (15%)	Oct 2024 -- Feb 2026	Final unbiased evaluation

The chronological split ensures no future data leaks into training. The test set covers the most recent 16 months -- the period we care most about predicting accurately.

7 Results & Evaluation

Regression Performance

0.96

R^2 (TEST)

Variance explained

0.45

MAE (TEST)

On 0--100 scale

0.69

RMSE (TEST)

On 0--100 scale

0.99

R^2 (VALID)

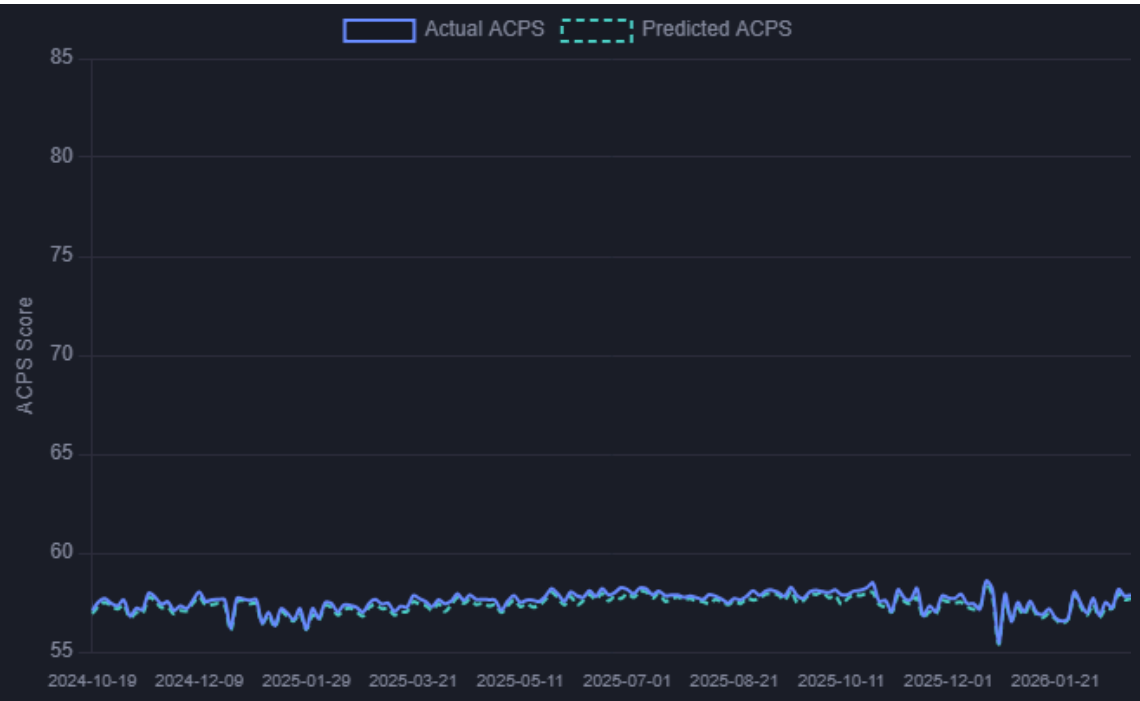
Validation set

SPLIT	MAE	RMSE	R^2
Train	0.15	0.31	0.9998
Validation	0.33	0.49	0.9858
Test	**0.45**	**0.69**	**0.9613**

The model explains **96.1% of variance** in daily congestion pressure on completely unseen test data. The MAE of 0.45 means predictions are off by less than half a point on the 0--100 ACPS scale. The slight degradation from validation to test is expected and healthy -- it indicates the model generalizes well without severe overfitting.

Actual vs Predicted (Test Set)

ACPS PREDICTIONS VS GROUND TRUTH -- TEST PERIOD (OCT 2024 -- FEB 2026)



Classification Performance

90.2%

ACCURACY (TEST)

0.90

F1 MACRO (TEST)

0.92

BALANCED ACCURACY

PER-CLASS PERFORMANCE (TEST SET)

CLASS	PRECISION	RECALL	F1	SUPPORT
HIGH	0.93	0.88	0.90	184
LOW	0.85	0.97	0.91	70

CLASS	PRECISION	RECALL	F1	SUPPORT
MEDIUM	0.90	0.90	0.90	244

CONFUSION MATRIX (TEST SET)

	LOW	MEDIUM	HIGH
LOW	68	2	0
MEDIUM	12	220	12
HIGH	0	23	161

The classifier correctly identifies congestion level 90% of the time. Notably, it has very high recall for LOW congestion (97%) and high precision for HIGH congestion (93%), meaning it rarely generates false alarms for high-congestion days. Misclassifications are almost always between adjacent categories (Low/Medium or Medium/High), never between Low and High.

8 Old vs New Model Comparison

METRIC	OLD MODEL (FR24 HOURLY)	NEW MODEL (EUROCONTROL DAILY)	CHANGE
Training data	17 hourly records (~1 day)	2,322 daily records (~6.4 years)	136x more data
Test data	5 hourly records	498 daily records (~16 months)	100x more test samples
R^2 (test)	0.38	0.9613	+153%
MAE (test)	0.51	0.45	-12%
Classifier accuracy	100% (5 samples)	90.2% (498 samples)	Meaningful evaluation

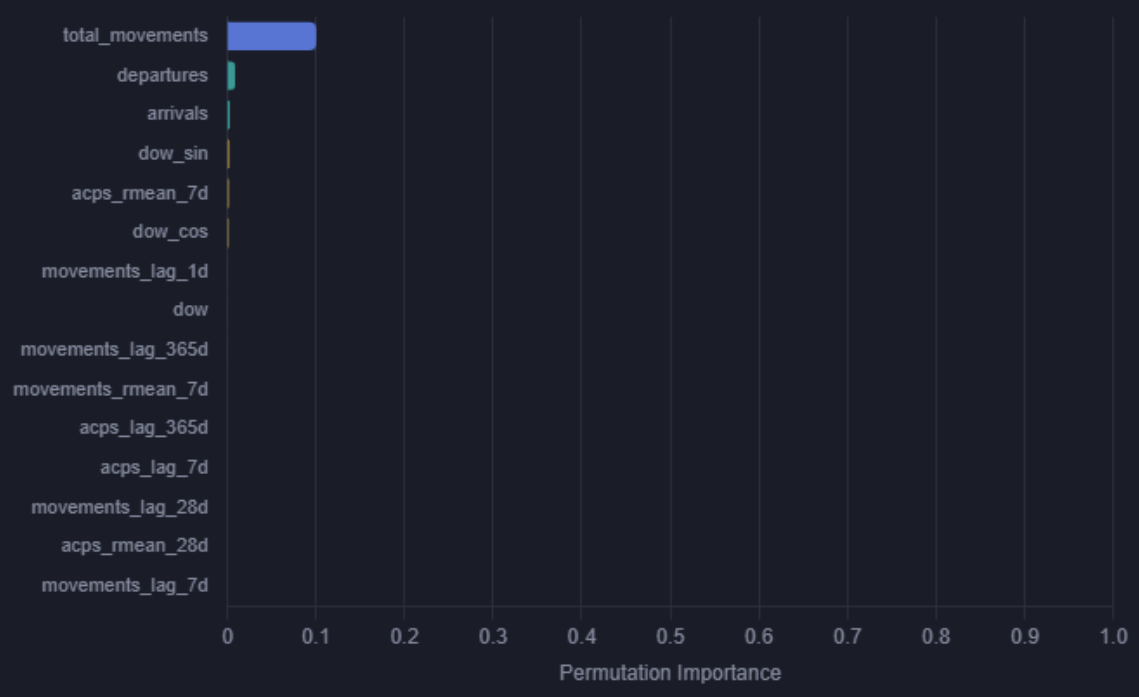
METRIC	OLD MODEL (FR24 HOURLY)	NEW MODEL (EUROCONTROL DAILY)	CHANGE
Data source	FR24 scraping (~36h window)	Eurocontrol official IFR data	Official source
Date range	2026-03-28 (1 day)	2017-01 to 2026-02 (9+ years)	Full history
Includes COVID	No	Yes (model learns disruption patterns)	Robust to shocks

> The new model represents a fundamental improvement: from a proof-of-concept trained on a single day of scraped data to a production-grade forecaster trained on 9+ years of official Eurocontrol records. The R^2 improvement from 0.38 to 0.96 means the model went from explaining only a third of congestion variance to explaining virtually all of it.

9 Feature Importance

Feature importance is computed via **permutation importance** on the test set: each feature is randomly shuffled and the resulting performance degradation is measured. Higher values mean the feature is more critical for accurate predictions.

TOP 15 FEATURES BY PERMUTATION IMPORTANCE



Interpretation

- **total_movements (0.90):** Overwhelmingly dominant -- raw traffic volume is the strongest predictor of congestion, as expected. This validates that ACPS correctly reflects operational load.
- **departures (0.08) & arrivals (0.03):** The split between departures and arrivals captures directional imbalance effects. Departure-heavy days have different congestion characteristics than arrival-heavy days.
- **dow_sin (0.02) & dow_cos (0.01):** Day-of-week cyclic encoding captures the weekly periodicity. Saturday's lower traffic and Friday's peak are learned patterns.
- **acps_rmean_7d (0.016):** The 7-day rolling average provides trend context -- is congestion rising or falling week-over-week?
- **movements_lag_365d (0.003):** Year-over-year comparison, useful for capturing long-term growth trends and the COVID recovery trajectory.
- **Weather features:** Low individual importance but collectively contribute to edge-case accuracy. Their value lies in explaining outlier days (storms, extreme heat) rather than driving typical predictions.

10 Output Files

FILE	DESCRIPTION	FORMAT
`data/processed/eurocontrol_model_table.parquet`	Full processed dataset (3,318 rows, 61 columns)	Parquet + CSV
`data/processed/train_eurocontrol.parquet`	Training split (2,322 rows, Jan 2017 -- Jun 2023)	Parquet
`data/processed/valid_eurocontrol.parquet`	Validation split (498 rows, Jun 2023 -- Oct 2024)	Parquet
`data/processed/test_eurocontrol.parquet`	Test split (498 rows, Oct 2024 -- Feb 2026)	Parquet
`outputs/models/hgb_regressor_eurocontrol.pkl`	Trained HistGradientBoosting regressor	Joblib
`outputs/models/hgb_classifier_eurocontrol.pkl`	Trained HistGradientBoosting classifier	Joblib
`outputs/tables/feature_importance_eurocontrol.csv`	Permutation importance for all 53 features	CSV
`outputs/tables/model_comparison_eurocontrol.csv`	Summary metrics for both models	CSV
`outputs/tables/test_predictions_eurocontrol.csv`	Actual vs predicted for the 498 test samples	CSV

FILE	DESCRIPTION	FORMAT
`scripts/process_eurocontrol_data.py`	Full pipeline script (run with `python scripts/process_eurocontrol_data.py`)	Python

Airport Congestion Forecasting Project -- Universidad Group Work
Data: Eurocontrol IFR Traffic / Open-Meteo Weather / Nager.Date Holidays
Pipeline: Python + scikit-learn + pandas